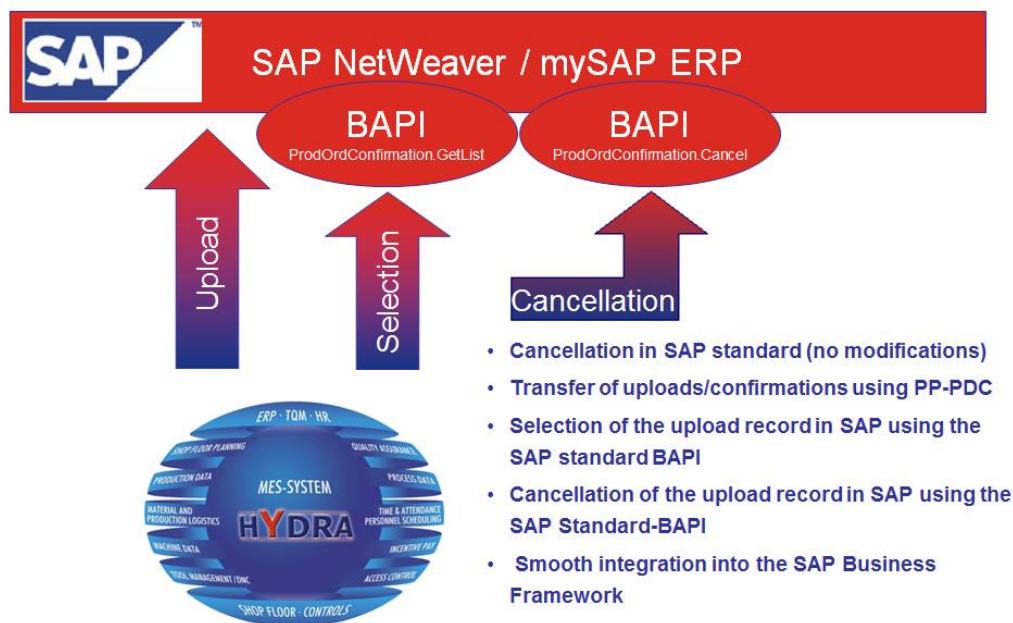


1 Cancellation Process

Technical basics of cancellation

Technically, uploads/confirmations are cancelled in SAP in two basic steps:

- Selecting the original data record to identify the SAP confirmation counter.
- Cancelling the original record using the confirmation counter.



The order key (order / sequence / operation / suboperation) and a unique ID (EX_IDENT) are required to select the confirmation counter. HYDRA generates this ID for each time ticket (original record) transferred to SAP. HYDRA sends this ID along with the time ticket. SAP keeps this ID with the operation confirmation/upload.

You can use the identified confirmation counter to clearly identify the time ticket (original record) and to cancel it in SAP.

Transferring cancellation records to SAP

In case of production orders, the postings recorded in HYDRA are promptly transferred to SAP and also posted there. If you change data subsequently in HYDRA, the same changes have to be repeated in SAP in order to avoid an imbalance between the data of both systems.

These are the steps required to execute and monitor the cancellation process:

Editing data in the HYDRA maintenance of postings

The following conditions must be met to transfer cancellations to SAP:

- The editing function "HYDRA maintenance of postings" is used.
- The order type is subject to confirmations/uploads.
- The order type allows for the posting to be changed once it has been uploaded.



The cancellation interface has not been released for use with the HYDRA event maintenance.

The system only generates cancellation records if the original data record has already been uploaded to SAP. The *Confirmed* column of the maintenance of postings indicates if a data record has already been uploaded to SAP.

If you change and save a data record in the maintenance of postings, only the current data record will be shown. The system still includes the original data record but does not show it.

Provision of data for the transfer to SAP

Time tickets are provided by converting the HYDRA data format into a meta format and transferring the data records to the HYDRA interface tables. The upload program generates a unique ID (EX_IDENT) during the transfer. This ID identifies each single record during the cancellation process.

If you use several HYDRA servers that upload data to an SAP instance, the EX_IDENT must differ from system to system to ensure uniqueness. To do so, the upload program (myerprck.exe/out) provides a parameter that guarantees this uniqueness. Use the following parameter to start the upload program:

/IDENT_PRAEFIX=XXX

Note: The prefix must **ONLY** include hexadecimal characters, i.e. 0-9 and A-F.

Displaying the records in MLE outbound transactions

After confirming the data records, they can be transferred to SAP in a meta format. The *Confirmed* column of the maintenance of postings indicates such data records using the icon .

In the MLE outbound transactions (menu: File --> System information --> MLE communication --> Outbound transactions), you can identify the uploads to SAP via the PP-PDC interface and the cancellation records as follows:

	Time ticket uploads	Cancellation records
Message type	PPCC2PRETTICKET	E2BP_PP_TIMETICKET_C
IDoc type	PPCC2PRETTICKET01	PPCC2PRETTICKET01
Segment type	E2BP_PP_TIMETICKET	E2BP_PP_TIMETICKET_C

Segment E2BP_PP_TIMETICKET_C

The segment of cancellation records E2BP_PP_TIMETICKET_C is structured as follows:

Field name	T	L	D	Description	Usage in HYDRA
CONF_NO	NUMC	10	0	Upload number of the operation	Upload number of the operation
ORDERID	CHAR	12	0	Order	According to configuration (*1)
SEQUENCE	CHAR	6	0	Sequence	According to configuration (*1)
OPERATION	CHAR	4	0	Operation	According to configuration (*1)
SUB_OPER	CHAR	4	0	Suboperation	According to configuration (*1)
PROC_START_TIME	TIME	6		Time when "starting processing"	Not used
EX_IDENT	CHAR	32	0	Unique reference to data record	Combined key: transfer parameter + ade_protokoll.verweis
POSTG_DATE	DATS	8	0	External date of entering the confirmation/upload	Depending on the program parameter /SDAT_STORNO of the program myerprck.exe/out : Not set (by default) : Change date of the log record /SDAT_STORNO set : Shift date of the log record
CONF_TEXT	CHAR	40		Confirmation/upload text for the cancellation record.	Reserved for future developments.

[The documentation of the standard interface](#) describes the structure of the standard records of the PP-PDC interface.

Process of transferring data to SAP

The SAP upload request controls the data transfer to SAP. The HYDRA MLE distribution model starts the HYDRA upload client. This upload client either transfers the time tickets to SAP using IDoc and tRFC or cancels the cancellation records via sRFC in SAP.

Initially, the upload client is called for the confirmation/upload of time tickets. The provision dates of the individual data records result in a sorting, which must be adhered to for serialization purposes.

In a first step, the upload client attempts to transfer time tickets to SAP. If this is possible, the upload client selects all time ticket records up to the first cancellation record and transfers the selected time tickets to SAP via IDoc. If it is not possible to select time tickets, the upload client checks whether there are cancellation records that have to be transferred to SAP.

If the upload client finds a cancellation record it will try to cancel it in SAP. Since cancellations are made using synchronous RFCs, you can state immediately whether a cancellation was successful.

If the cancellation process was successful the upload client determines the next records ready to be transferred (time tickets or cancellation records) based on the chronological order of their provision and transfers them to SAP.

If the cancellation was not successful, the cancellation record will be set to "TODO" in the interface. Consequently, the record is registered for the next transfer. In this confirmation/upload cycle, no additional data records will be transferred to SAP that were provided after this record.

The number of posting attempts is managed for the data record in order to prevent the confirmation/upload interface from being completely blocked for this message type. The upload client provides a parameter that allows to define after how many attempts a data record will be qualified as incorrect and when to proceed with the "normal" uploads.

The parameter `"/REDO_CANCEL=` of the program `hysapupl.exe` is included in the HYDRA MLE distribution model and can be changed at any time.

The SAP upload request triggers the next transfer cycle, therefore this request should be transferred to HYDRA at intervals of 5 minutes.

By default, the upload client cancels one posting record in SAP per posting record in HYDRA. There might be specific, customer-related situations where the system generates several SAP posting records per HYDRA posting record.

In order to handle this situation, you can set the parameter `"/MULTIPLE_CANCEL` of the `hysapupl.exe` program in the HYDRA MLE distribution model. This will cancel all SAP posting records existing for a HYDRA posting record.

SAP operation status

HYDRA shows the same behavior when canceling data via the interface as SAP when collecting data via SAP-GUI. If a partial confirmation/upload is canceled for already completed operations and if this is entered again, SAP will set the operation status from finally confirmed to partially confirmed.

After correction of a "Posting/ Partial confirmation" (L20) in HYDRA, HYDRA will transfer the new "Posting/ Partial confirmation" also as record type L20 to SAP, irrespective of the operation status in HYDRA. If a final confirmation (L40) has been corrected in HYDRA, HYDRA will transfer this new final confirmation as record type L40 to SAP.